

Surface Integrals

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18 Surface Integrals

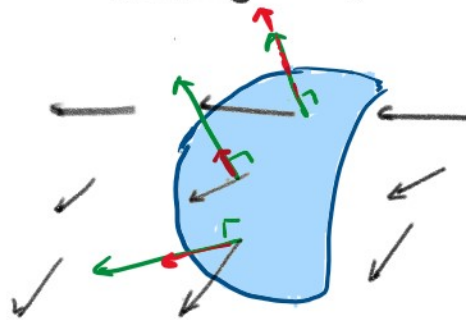
18.1 For surface S : $\vec{r}(u,v) = [x(u,v), y(u,v), z(u,v)]$, $(u,v) \in R$

with $\vec{N} = \vec{r}_u \times \vec{r}_v$ and $\vec{n} = \frac{1}{|\vec{N}|} \vec{N}$

and a vectorfield $\vec{F} = [F_1, F_2, F_3]$ defined on domain D containing S ,
we define the surface integral:

$$\iint_S \vec{F} \cdot \vec{n} \, dS = \iint_R \vec{F}(\vec{r}(u,v)) \cdot \vec{N}(u,v) \, du \, dv$$

We call this the flux integral of $\vec{F}$ through S as it accumulates
the normal component of
 $\vec{F}$ across the surface S .



18.2

EXAMPLE 1 Flux Through a Surface

Compute the flux of water through the parabolic cylinder $S: y = x^2, 0 \leq x \leq 2, 0 \leq z \leq 3$ (Fig. 245) if the velocity vector is $\mathbf{v} = \mathbf{F} = [3z^2, 6, 6xz]$, speed being measured in meters/sec. (Generally, $\mathbf{F} = \rho \mathbf{v}$, but water has the density $\rho = 1 \text{ g/cm}^3 = 1 \text{ ton/m}^3$.)

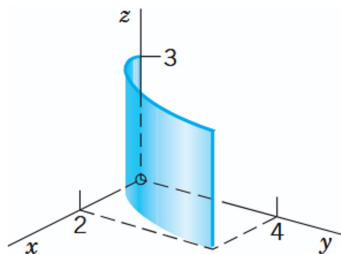


Fig. 245. Surface S in Example 1

Solution. Writing $x = u$ and $z = v$, we have $y = x^2 = u^2$. Hence a representation of S is

$$S: \quad \mathbf{r} = [u, u^2, v] \quad (0 \leq u \leq 2, 0 \leq v \leq 3).$$

By differentiation and by the definition of the cross product,

$$\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v = [1, 2u, 0] \times [0, 0, 1] = [2u, -1, 0].$$

On S , writing simply $\mathbf{F}(S)$ for $\mathbf{F}[\mathbf{r}(u, v)]$, we have $\mathbf{F}(S) = [3v^2, 6, 6uv]$. Hence $\mathbf{F}(S) \cdot \mathbf{N} = 6uv^2 - 6$. By integration we thus get from (3) the flux

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, dA &= \int_0^3 \int_0^2 (6uv^2 - 6) \, du \, dv = \int_0^3 (3u^2v^2 - 6u) \Big|_{u=0}^2 \, dv \\ &= \int_0^3 (12v^2 - 12) \, dv = (4v^3 - 12v) \Big|_{v=0}^3 = 108 - 36 = 72 \text{ [m}^3/\text{sec]} \end{aligned}$$

or 72,000 liters/sec. Note that the y -component of $\mathbf{F}$ is positive (equal to 6), so that in Fig. 245 the flow goes from left to right.

EXAMPLE 2 Surface Integral

Evaluate (3) when $\mathbf{F} = [x^2, 0, 3y^2]$ and S is the portion of the plane $x + y + z = 1$ in the first octant (Fig. 246).

Solution. Writing $x = u$ and $y = v$, we have $z = 1 - x - y = 1 - u - v$. Hence we can represent the plane $x + y + z = 1$ in the form $\mathbf{r}(u, v) = [u, v, 1 - u - v]$. We obtain the first-octant portion S of this plane by restricting $x = u$ and $y = v$ to the projection R of S in the xy -plane. R is the triangle bounded by the two coordinate axes and the straight line $x + y = 1$, obtained from $x + y + z = 1$ by setting $z = 0$. Thus $0 \leq x \leq 1 - y, 0 \leq y \leq 1$.

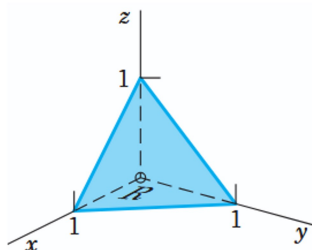


Fig. 246. Portion of a plane in Example 2

By inspection or by differentiation,

$$\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v = [1, 0, -1] \times [0, 1, -1] = [1, 1, 1].$$

Hence $\mathbf{F}(S) \cdot \mathbf{N} = [u^2, 0, 3v^2] \cdot [1, 1, 1] = u^2 + 3v^2$. By (3),

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, dA &= \iint_R (u^2 + 3v^2) \, du \, dv = \int_0^1 \int_0^{1-v} (u^2 + 3v^2) \, du \, dv \\ &= \int_0^1 \left[\frac{1}{3}(1-v)^3 + 3v^2(1-v) \right] dv = \frac{1}{3}. \end{aligned}$$

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18.3

EXAMPLE 3 Change of Orientation in a Surface Integral

In Example 1 we now represent S by $\tilde{\mathbf{r}} = [v, v^2, u]$, $0 \leq v \leq 2$, $0 \leq u \leq 3$. Then

$$\tilde{\mathbf{N}} = \tilde{\mathbf{r}}_u \times \tilde{\mathbf{r}}_v = [0, 0, 1] \times [1, 2v, 0] = [-2v, 1, 0].$$

For $\mathbf{F} = [3z^2, 6, 6xz]$ we now get $\tilde{\mathbf{F}}(S) = [3u^2, 6, 6uv]$. Hence $\tilde{\mathbf{F}}(S) \cdot \tilde{\mathbf{N}} = -6u^2v + 6$ and integration gives the old result times -1 ,

$$\iint_R \tilde{\mathbf{F}}(S) \cdot \tilde{\mathbf{N}} \, dv \, du = \int_0^3 \int_0^2 (-6u^2v + 6) \, dv \, du = \int_0^3 (-12u^2 + 12) \, du = -72.$$

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19 Surface Integral of Scalar Field

19.1 Let $G(\mathbf{r})$ be a scalar function. We define the surface integral

$$\iint_S G(\mathbf{r}) \, dA = \iint_R G(\mathbf{r}(u, v)) |\mathbf{N}(u, v)| \, du \, dv.$$

19.2

As for applications, if $G(\mathbf{r})$ is the mass density of S , then (6) is the total mass of S . If $G = 1$, then (6) gives the **area** $A(S)$ of S ,

$$(8) \quad A(S) = \iint_S dA = \iint_R |\mathbf{r}_u \times \mathbf{r}_v| \, du \, dv.$$

EXAMPLE 4 Area of a Sphere

For a sphere $\mathbf{r}(u, v) = [a \cos v \cos u, a \cos v \sin u, a \sin v]$, $0 \leq u \leq 2\pi$, $-\pi/2 \leq v \leq \pi/2$ [see (3) in Sec. 10.5], we obtain by direct calculation (verify!)

$$\mathbf{r}_u \times \mathbf{r}_v = [a^2 \cos^2 v \cos u, a^2 \cos^2 v \sin u, a^2 \cos v \sin v].$$

Using $\cos^2 u + \sin^2 u = 1$ and then $\cos^2 v + \sin^2 v = 1$, we obtain

$$|\mathbf{r}_u \times \mathbf{r}_v| = a^2(\cos^4 v \cos^2 u + \cos^4 v \sin^2 u + \cos^2 v \sin^2 v)^{1/2} = a^2 |\cos v|.$$

With this, (8) gives the familiar formula (note that $|\cos v| = \cos v$ when $-\pi/2 \leq v \leq \pi/2$)

$$A(S) = a^2 \int_{-\pi/2}^{\pi/2} \int_0^{2\pi} |\cos v| du dv = 2\pi a^2 \int_{-\pi/2}^{\pi/2} \cos v dv = 4\pi a^2.$$

Representations $z = f(x, y)$. If a surface S is given by $z = f(x, y)$, then setting $u = x$, $v = y$, $\mathbf{r} = [u, v, f]$ gives

$$|\mathbf{N}| = |\mathbf{r}_u \times \mathbf{r}_v| = |[1, 0, f_u] \times [0, 1, f_v]| = |[-f_u, -f_v, 1]| = \sqrt{1 + f_u^2 + f_v^2}$$

and, since $f_u = f_x, f_v = f_y$, formula (6) becomes

$$(11) \quad \iint_S G(\mathbf{r}) dA = \iint_{R^*} G(x, y, f(x, y)) \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2} dx dy.$$

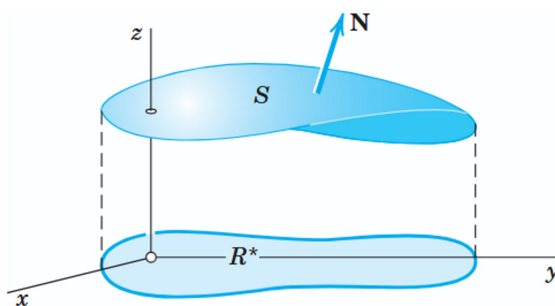


Fig. 250. Formula (11)

Here R^* is the projection of S into the xy -plane (Fig. 250) and the normal vector $\mathbf{N}$ on S points *up*. If it points *down*, the integral on the right is preceded by a minus sign.

From (11) with $G = 1$ we obtain for the **area** $A(S)$ of $S: z = f(x, y)$ the formula

$$A(S) = \iint_{R^*} \sqrt{1 + f_x^2 + f_y^2} dx dy$$

(12)

$$A(S) = \int \int_{R^*} \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2} dx dy$$

where R^* is the projection of S into the xy -plane, as before.